ARTIFICIAL INTELLIGENCE

Assignment: Logical Reasoning, Semantic Networks and NLP

TOTAL: 10 MARKS

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Instructions

- Answer all questions in your own words.
- Show your reasoning, not just the final answer.
- Draw the semantic network neatly by hand or using a digital tool.
- Complete the small Python coding activities.
- Include the code and its output in your submission.
- You may use VS Code or any other platform to execute Python code.

Part A — Logical reasoning (3 marks)

Q1. Reasoning about airline messages (3 marks)

An airline uses these rules:

- If a tweet reports a cancelled or delayed flight, it is a service complaint.
- If a tweet is a service complaint, it should be prioritised for support.
- If a tweet only asks for general flight information, it is an information request.

Classify the following messages:

A. “My flight has been cancelled and nobody has contacted me.”

B. “What time does flight AB123 depart?”

a) State the category of Tweet A and Tweet B. (1 mark)

Tweet A – Service Complaint

Tweet B – Information Request

b) Which tweet should be prioritised? Explain the rule-based reasoning step by step. Can you conclude that Tweet B is positive or negative? Explain why. (1 mark)

Tweet A should be prioritised. Rule 1 states that if a tweet reports a cancelled flight, it is a service complaint. Tweet A contains the word "cancelled", so it considered a service complaint. Rule 2 states that service complaints should be prioritised for support. Therefore, Tweet A gets priority. We cannot conclude if Tweet B is positive or negative. It is just asking for a departure time making it is a neutral request for information.

c) Complete the following Python code to implement the decision rules. (1 mark)

```
def classify_message(tweet):
tweet = tweet.lower()

if "cancelled" in tweet or "delayed" in tweet:
return "Service Complaint"

elif "what time" in tweet or "departure" in tweet:
return "Information Request"

else:
return "Other"

tweet_a = "My flight has been cancelled"
tweet_b = "What time does flight AB123 depart?"

print(classify_message(tweet_a))
print(classify_message(tweet_b))
```

Explain the output produced by the code.

Output:

Service Complaint

Information Request

The code first converts the text to lowercase. It checks Tweet A for the words "cancelled" or "delayed", finds "cancelled", and returns the string "Service Complaint". For Tweet B, it finds the phrase "what time" and returns "Information Request".

Part B — Semantic network (3 marks)

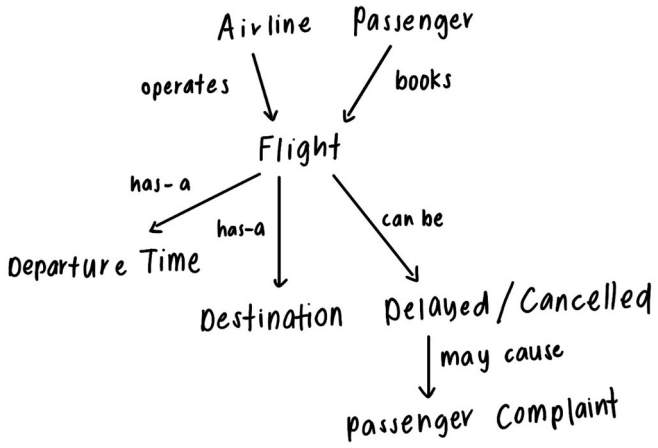
Q2. Representing airline knowledge (3 marks)

Create a semantic network using the following facts:

- An Airline operates Flights.
- A Flight has a Departure Time and a Destination.
- A Passenger books a Flight.
- A Flight can be Delayed or Cancelled.
- A Delay may cause a Passenger Complaint.

Total: 10 marks

a) Draw a semantic network containing the concepts and labelled relationships above. (2 marks)



b) Represent at least four relationships from your semantic network using Python. (1 mark)

Use the following example as a starting point:

```
semantic_network = [
    ("Airline", "operates", "Flight"),
    ("Passenger", "books", "Flight"),
    ("Flight", "has", "Destination"),
    ("Flight", "has", "Departure Time")
]

for relationship in semantic_network:
    print(relationship)
```

Expected output:

```
('Airline', 'operates', 'Flight')

('Passenger', 'books', 'Flight')

('Flight', 'has', 'Destination')

('Flight', 'has', 'Departure Time')
```

Add at least one more relationship of your own.

semantic_network = [

("Airline", "operates", "Flight"),

("Passenger", "books", "Flight"),

("Flight", "has", "Destination"),

("Flight", "has", "Departure Time"),

Total: 10 marks

```
    ("Flight", "can be", "Delayed"),
    ("Flight", "can be", "Cancelled"),
    ("Delay", "may cause", "Passenger Complaint")
]
for relationship in semantic_network:
    print(relationship)
```

Part C — Natural Language Processing (4 marks)

Q3. Text preprocessing and sentiment (2 marks)

Consider this tweet:

@SkyAir My flight is NOT good! It was delayed again. https://airline.example

a) Replace the username with USER and the URL with URL, convert the text to lowercase, and remove extra whitespace. Write the resulting text. (1 mark)

user my flight is not good! it was delayed again. url

b) Complete the following Python code to perform basic text preprocessing. (1 mark)

```
import re

def preprocess_text(text):

    # Replace usernames with USER
    text = re.sub(r"@\w+", "USER", text)

    # Replace URLs with URL
    text = re.sub(r"https?://\S+", "URL", text)

    # Convert text to lowercase
    text = text.lower()

    # Remove extra whitespace
    text = re.sub(r"\s+", " ", text).strip()

    return text


tweet = "@SkyAir My flight is NOT good! It was delayed again.
https://airline.example"

clean_tweet = preprocess_text(tweet)

print(clean_tweet)
```

Total: 10 marks

Write the output and explain why the word "not" should not be removed during preprocessing.

user my flight is not good! it was delayed again. url

The word "not" should never be removed because it is a negation word that completely flips the meaning of a sentence. If we remove it, "my flight is not good" becomes "my flight is good" which change from negative to positive sentence.

Q4. Evaluating a sentiment classifier (2 marks)

A classifier correctly predicts 80 out of 100 tweets, but it fails to identify most neutral tweets.

a) Is 80% accuracy alone enough to conclude that the classifier performs well? Explain. (1 mark)

No, 80% accuracy alone is not enough. If the dataset is unbalanced (for example, 80% of the tweets are positive/negative and only 20% are neutral), the classifier could just guess the majority class every time and still score 80%, while completely failing to identify the neutral tweets.

b) Name one metric that gives a more balanced view across negative, neutral and positive classes. Explain why it is useful. (1 mark)

The F1-score gives a more balanced view. It is useful because it calculates the balance between precision and recall for every class individually.